Justin Cui

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Research Interests

Human-AI Interaction · Intelligent User Interfaces · Human-AI Co-Creation · Personalization · Large Language Models

Education

Carnegie Mellon University, School of Computer Science 08/2026 – Present
Master of Intelligent Information System, Language Technologies Institute *Pittsburgh, PA*

University of Toronto 09/2021 – 04/2026
BASc, Engineering Science – Machine Intelligence *Toronto, ON, CA*
Major GPA: ***3.97****/4.0*

Research Experience

Undergraduate Researcher Aug 2025 – Present
DGP Lab, University of Toronto (Advisor: Prof. Tovi Grossman) *Toronto, ON, CA*

- Developed **LOOM**, a proactive system that transforms transient LLM chats into cumulative knowledge via a **dynamic learner memory graph**; implemented an inference pipeline that detects evolving goals to sequence scaffolded lessons.
- Designed an **interactive topology visualization** that externalizes the "knowledge space" to expose structural gaps, creating a **mixed-initiative** workflow where users actively prune or expand branches to refine their mental models.
- Conducted a formative **user study** (N=10) demonstrating how visualizing the learning solution space helps users refine vague intent and navigate non-linear knowledge gaps ("unknown unknowns"); work accepted to **PerFM @ AAAI 2026**.

Undergraduate Researcher Aug 2025 – Present
Vector Institute (Advisor: Prof. Zhijing Jin) *Toronto, ON, CA*

- Led the design of a **causal DataAgent** that, given a natural language causal question, searches **Harvard Dataverse** and ranks datasets suitable for the intended estimation task; applied causal estimation specific query reformulation with dense retrieval and multi field sparse matching to improve relevance.
- Co-curated **CauSciBench**, a benchmark of 367 tasks from 100+ papers across nine disciplines; evaluated **end-to-end causal pipelines** from problem formulation to modeling and interpretation, showing the best model still had ∼49% **mean relative error** on real-paper tasks.

Undergraduate Researcher Apr 2025 – Aug 2025
Vector Institute (Advisor: Prof. Alán Aspuru-Guzik) *Toronto, ON, CA*

- Investigated **compositional diffusion models** for the Abstraction and Reasoning Corpus (ARC), proposing a framework that decouples geometric occupancy from color pattern synthesis to improve few-shot generalization.
- Conducted a systematic **ablation study** on diffusion objectives (predicting noise ϵ vs. signal x_0) and loss landscapes (MSE vs. BCE); demonstrated that direct signal reconstruction significantly stabilizes convergence for sparse, discrete 2D grid tasks.

Undergraduate Researcher Apr 2023 – Aug 2025
Data-Driven Decision Making Lab, University of Toronto (Advisor: Prof. Scott Sanner) *Toronto, ON, CA*

- Co-led the development of **RA-Rec**, one of the first open-source **RAG-based conversational recommenders**; replaced rigid slot-filling with **semi-structured dialogue state tracking** to capture complex, evolving user intents across multi-turn interactions. Work accepted at **SIGIR 2024.**
- Engineered the retrieval pipeline using **two-tower dense retrieval** to match user states against item reviews, implementing **late fusion** to aggregate review-level relevance into final item scores for high-precision ranking.
- Proposed an **elaborative query reformulation** method to bridge the semantic gap between user inputs and item descriptions, validating the approach on a curated **3-domain synthetic dataset** (released on **HuggingFace**) with consistent **MAP@K** gains.

Industry Experience

Machine Learning Engineer Intern May 2024 – Apr 2025
Modiface *Toronto, ON, CA*

- Designed and implemented a **Stable Diffusion** pipeline on **GCP** for 3D facial dataset augmentation, processing 25M synthetic images and improving facial landmark prediction accuracy by **17%**.

- Optimized GenAI inference latency by 40%+ via **OpenVINO** integration for GAN/diffusion models.
- Built a supervised fine-tuning pipeline on 5k+ curated dialogue pairs, reducing API cost by **83%** versus in-context prompting.

Software Engineer Intern May 2022 – Sep 2022
Voith Hydro *Montreal, QC, CA*

- Automated material-spec analysis via **Python** (**pandas**/NumPy) to process CSV datasets and generate standardized reports, reducing manual review time.
- Built an internal infopoint centralizing resources for 150+ engineers; achieved a **95%** adoption rate in the first month.

Publications & Preprints

* Equal contribution

ChatWeave: Surfacing Latent Context Structure Across LLM Chat Conversations for Controllable and Transparent Personalization
Justin Cui, Kevin Pu, Tovi Grossman
Manuscript in preparation for CHI 2027.

LOOM: Personalized Learning Informed by Daily LLM Conversations Toward Long-Term Mastery via a Dynamic Learner Memory Graph
Justin Cui, Kevin Pu, Tovi Grossman
PerFM @ AAAI 2026. [paper] [github].

Retrieval-Augmented Conversational Recommendation with Prompt-based Semi-Structured Natural Language State Tracking
Sara Kemper*, **Justin Cui***, Kai Dicarlantonio*, Kathy Lin*, Danjie Tang*, Anton Korikov, Scott Sanner
ACM SIGIR 2024. [paper] [github].

Bayesian Active Learning with Gaussian Processes Guided by LLM Relevance Scoring for Dense Passage Retrieval
Junyoung Kim, Anton Korikov, Jiazhuo Liang, **Justin Cui**, Yifan Liu, Qianfeng Wen, Mark Zhao, Scott Sanner
Findings of ACL 2026, 9884–9898.

Multimodal Item Scoring for Natural Language Recommendation via Gaussian Process Regression with LLM Relevance Judgments
Yifan Liu*, Qianfeng Wen*, Jiazhuo Liang*, Mark Zhao*, **Justin Cui**, Anton Korikov, Armin Toroghi, Junyoung Kim, Scott Sanner
Findings of ACL 2026. [paper].

A Simple but Effective Elaborative Query Reformulation Approach for Natural Language Recommendation
Qianfeng Wen*, Yifan Liu*, **Justin Cui***, Joshua Zhang, Anton Korikov, George-Kirollos Saad, Scott Sanner
arXiv preprint 2025. [paper] [github].

CauSciBench: A Comprehensive Benchmark on End-to-End Causal Inference for Scientific Research
Sawal Acharya, Terry Jingchen Zhang, Andrew Kim, Anahita Haghighat, Sun Xianlin, Pepijn Cobben, Rahul Babu Shrestha, Maximilian Mordig, Jacob T. Emmerson, Furkan Danisman, Yuen Chen, Clijo Jose, Andrei Ioan Muresanu, **Justin Cui**, Jiarui Liu, Yahang Qi, Punya Syon Pandey, Yinya Huang, Bernhard Schölkopf, Mrinmaya Sachan, Zhijing Jin
ICLR 2026 [paper].

Honors & Awards

Engineering Science Research Opportunities Fellowship ($7,000) – Vector Institute Apr 2025
ESROP-Global Research Fellowship ($9,000) – ETH Zurich (Declined for Vector Inst.) Mar 2025
Wallberg Undergraduate Scholarship ($1,500) – University of Toronto Aug 2024
ESROP-Global Research Fellowship ($10,000) – TU Munich (Declined for D3M Lab) Apr 2023
NSERC Undergraduate Student Research Award (USRA) ($7,500) – NSERC / UofT Apr 2023
Wallberg Undergraduate Scholarship ($1,500) – University of Toronto Aug 2022
Fujino/Smith Emergence Scholarship ($1,500) – University of Toronto Mar 2022
Dean's Honours List – University of Toronto (Received every semester) 2021 – Present

Service & Leadership

Class Representative, EngSci Machine Intelligence Review Committee 2024 – Present
Represented student perspectives in the first major curriculum audit since the major's inception.

Reviewer, ACM SIGIR 2025 Feb 2025 – Mar 2025

Reviewed demo submissions for scientific quality, novelty, and reproducibility.

Academic Mentor, Engineering Science 2022 – Present

Mentored 1–2 lower-year students annually, providing guidance on academic navigation and student life.

Team Captain, UofT Engineering Intramurals Soccer 2022 – Present

Fostered team cohesion and student engagement within the engineering faculty, leading the team to an Intramural Championship.